

Figure 3: Results of entity alignment and relation alignment w.r.t the number of iterations.

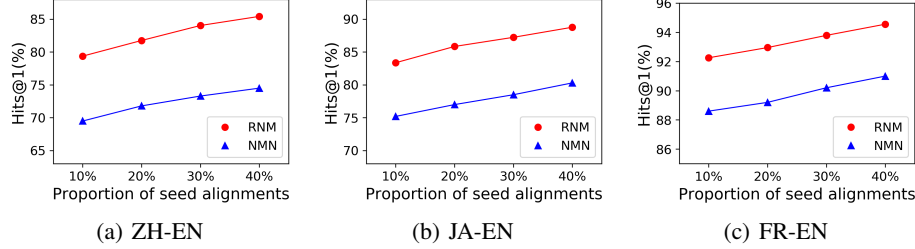


Figure 4: Results of entity alignment w.r.t the proportion of seed alignments.

Relation Alignment The proposed model RNM can not only be used for entity alignment but also for relation alignment. Table 4 shows the comparison results of different methods on the three datasets for relation alignment. Note that the results of baselines are from the reported data in (?), where -R denotes the original model for relation alignment, -PR denotes the model that approximates the relation representations using entity embeddings as in (?), and -JR denotes the model that jointly learns the embeddings of both entities and relations. From the results we can observe that the proposed model RNM performs better than all the baselines especially in Hits@1. Among the baselines, BootEA achieves better results compared with MTransE due to its bootstrapping strategy, while GCN further improves the performance by incorporating the semantic information. The proposed model RNM outperforms the best baseline model (HGCN-JR) by 10.3%, 9.5%, and 7.0% in Hits@1 on the three datasets, respectively. The reason may be that RNM aligns relations by matching the head and the tail entities which can provide more evidence for relation alignment. Moreover, these results confirm our assumption that the entity alignment and the relation alignment can reinforce each other in our model.

Analysis Figure 3 shows the results of entity alignment and relation alignment with different numbers of iterations for the proposed model RNM. With the increase of iterations, the performance of RNM on entity alignment and relation alignment raise accordingly. This proves the effectiveness of the iterative framework of RNM and the assumption that the entity alignment task and the relation alignment task can reinforce each other for better performance.

Since RNM can implement the alignments in a semi-supervised manner, we conduct several experiments on the

entity alignment task with different proportions of seed alignments, and results are shown in Figure 4. We choose NMN, which performs the best among baselines, as the comparison model. From the results, we can observe that RNM outperforms NMN in all situations. Even RNM with only 10% seed alignments performs better than the NMN with 40% seed alignments on all the three datasets. This is because RNM explores the useful information of relations when matching neighborhood, and the iterative strategy help to enhance the performance.

Conclusion and Future Work

In this paper, we propose a novel relation-aware neighborhood matching model named RNM for entity alignment. In the model, we jointly learn the embeddings of entities and relations. Moreover, we propose to make use of the semantic information and mapping properties of relations for better entity alignment. In addition, we implement entity alignment and relation alignment iteratively to reinforce each other in a semi-supervised manner. Finally, we evaluate the proposed model on three cross-lingual KG datasets and empirical results demonstrate the effectiveness of RNM.

In the future, we will study how to make use of the side information such as attributes (?) and descriptions (?) to improve the accuracy of entity alignment.

Acknowledgments

This work was supported by the National Key Research and Development Project (Grant No.2020AAA0106600) and Peking University Education Big Data Project (Grant No.2020YBC10).

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